BITS Pilani

# Automotive Vehicle

22-08-2026

Dr. Srinibas Tripathy
Department of Mechanical Engineering

innovate achieve lead

BITS Pilani

# Automotive Vehicle [AELZC441]
# Lecture No. 05: Chassis-body & Engines

# Recap of previous section

- Introduction
- Testing agencies for homologation (certification)
- virtual Engg.
- prod. dev.

# Sections to be covered

| Contact Session | List of Topic Title | Sub-Topics |
|---|---|---|
| 1 | Introduction to automobiles | - Automobiles, classification, and homologation<br>- Development methodology, virtual engineering methods |
| 2 | Chassis and Body | - Types of chassis and applications<br>- Weight distribution and analysis |
| 3 | Powertrain Fundamentals | - Vehicle propulsion system requirements<br>- Tractive force and torque requirement<br>- Powertrain architecture |
| 4,5,6 | IC Engine Fundamentals | - Fundamentals of IC Engine<br>- Fuel conversion techniques<br>- Torque characteristics<br>- Emissions in engines |

1 Sesn

1 in

2 Sessions

I HEVs, EVs

# Vehicle Body and Chassis

# Vehicle Body and Chassis

❖ **Vehicle Structure: Chassis + Body**

S+B

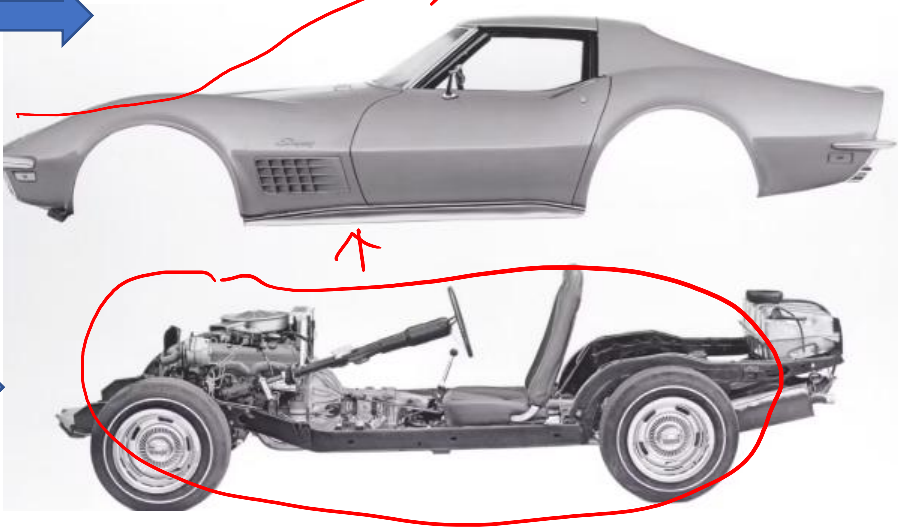
➢ **Body**

Structure enclosing the passenger/cargo space and providing shape, protection, aerodynamics and appearance.

object efficiency

➢ **Chassis**

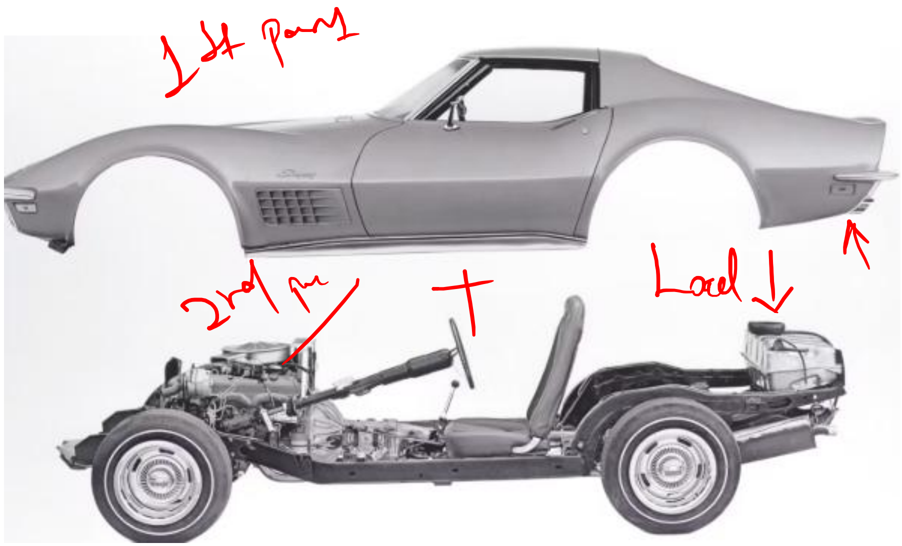
Structural skeleton with major mechanical systems (Suspension, steering, brakes, powertrain, wheels, etc.

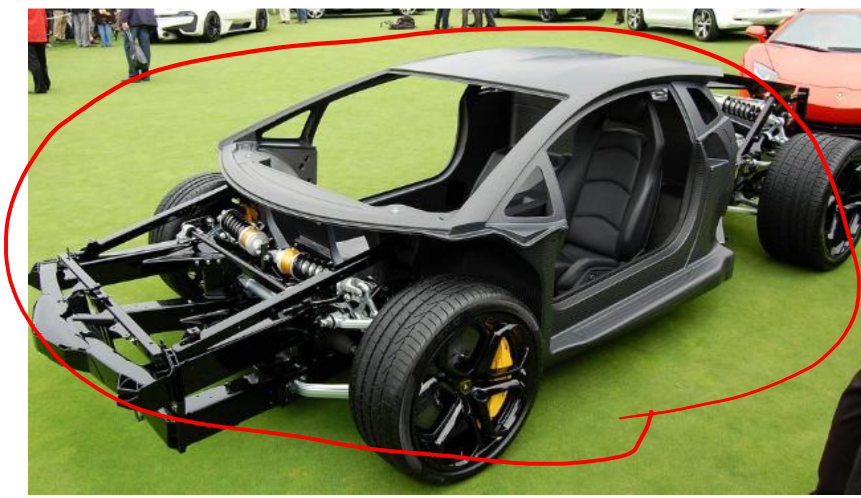


# Fundamental constructions

- **Body-on-Frame Construction**
  Body and load-carrying frame are separate structures.

- **Integral / Unibody Construction**
  Body and structural frame are integrated.



Vehicle Structure

# Body-on-Frame Vs Unibody

| Parameters | Body-on-Frame Construction | Unibody Construction |
|---|---|---|
| **Construction** | Body mounted on a structural frame | Body and chassis form a single integrated |
| **Load carrying** | By structural frame | By both Body and chassis |
| **Weight** | Higher | Lower |
| **Fuel Efficiency** | Lower | Better |
| **Towing / Payload** | Excellent | lower for comparable vehicle classes |
| **Repair** | Easily repaired/replaced | Expensive to repair |
| **Applications** | Trucks, pickups, and off-road vehicles | Passenger cars, crossovers, many SUVs |

- **Body-on-Frame: Strength + Payload/Towing + Durability**
- **Unibody: Lightweight + Handling + Efficiency**

# Unibody construction

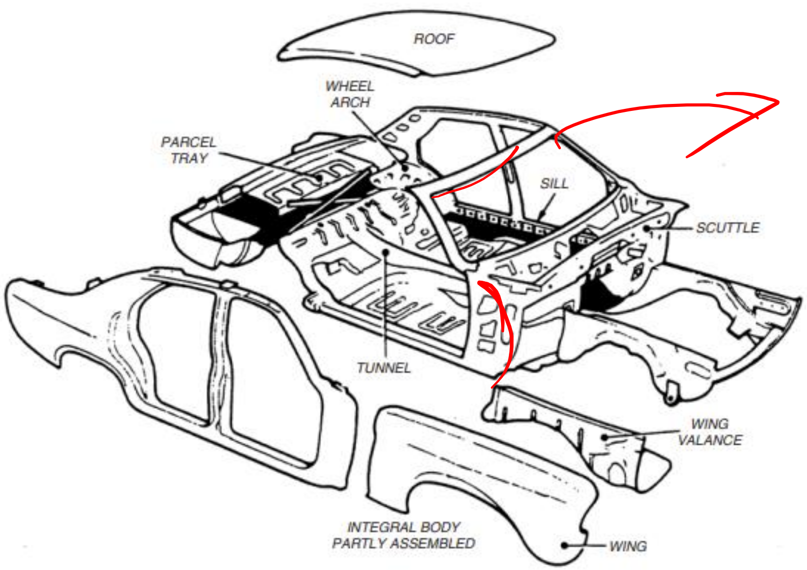


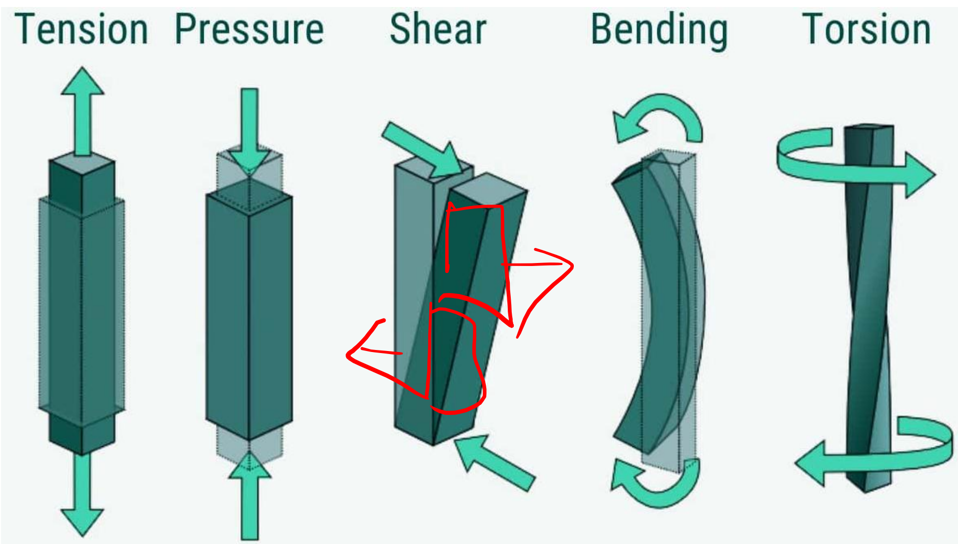

**Figure: Framelss/Integral body construction**

**Table: Thickness of different sections**

| Component / Application | Typical Thickness (mm) | Remarks |
|---|---|---|
| Large-area panels (roof, quarters, doors, and floors) | 0.915 | Subjected mainly to direct strain |
| Panels with curved contours | 0.765 – 0.66 | Curvature improves stiffness and rigidity |
| Major structural parts (cross members, rails, and pillars) | 1.00 – 1.25 | Load-carrying structural members |
| Local reinforcements | 1.625 | Used where additional strength is required |

0.66 - 1.625 mm

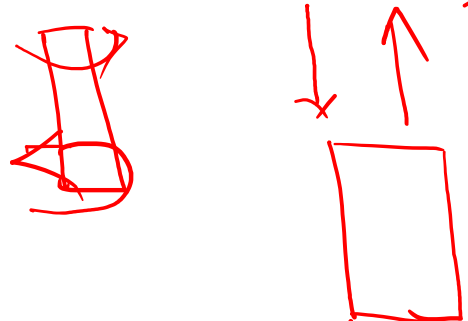
# Loading on Chassis




Pull Push twist

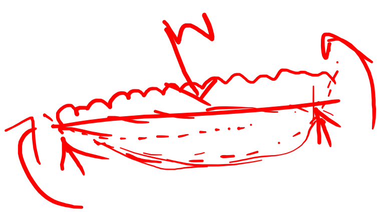
Compressive force

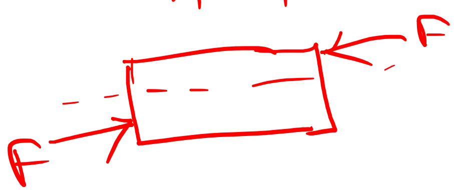
Tensile

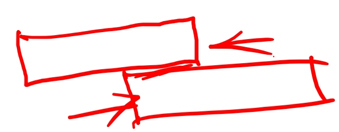


Pushing away

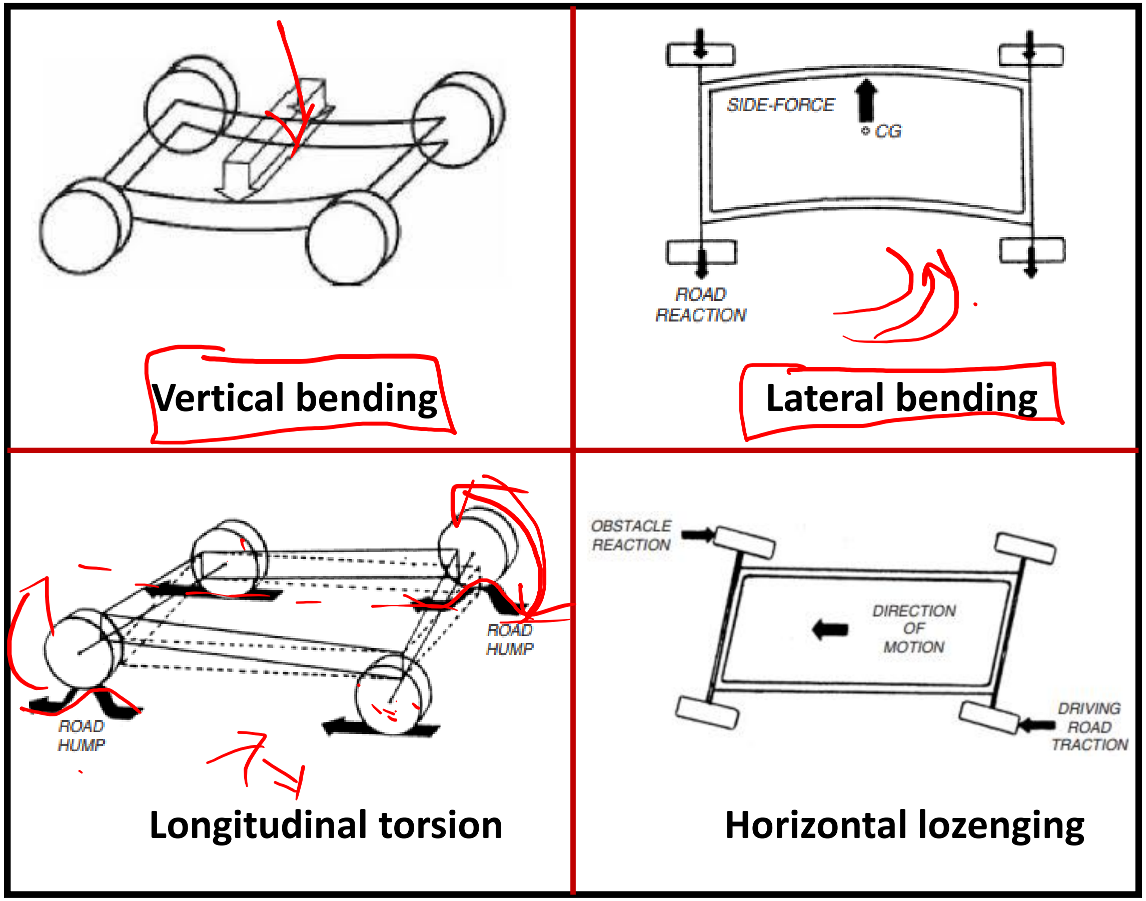
Pull







# Loading on Chassis




1. **Vertical bending:**

   Vehicle weight

2. **Lateral bending:**

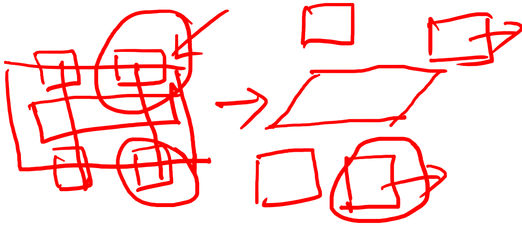
   Cornering forces

3. **Longitudinal torsion:**

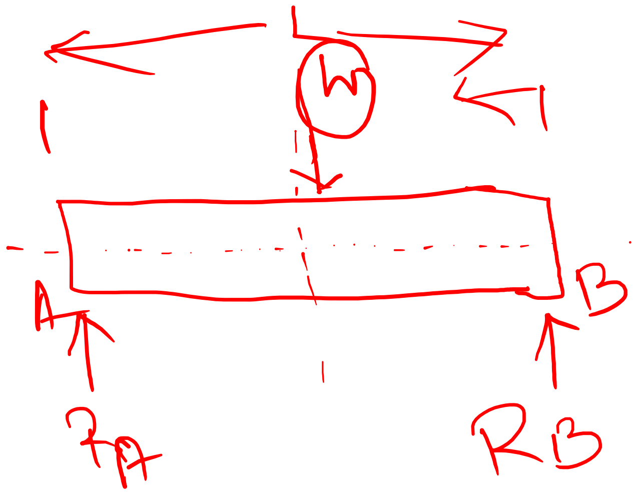
   Different road heights

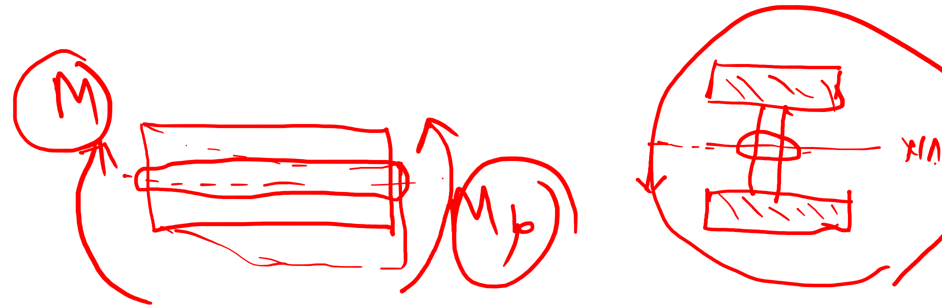
4. **Horizontal lozenging:**

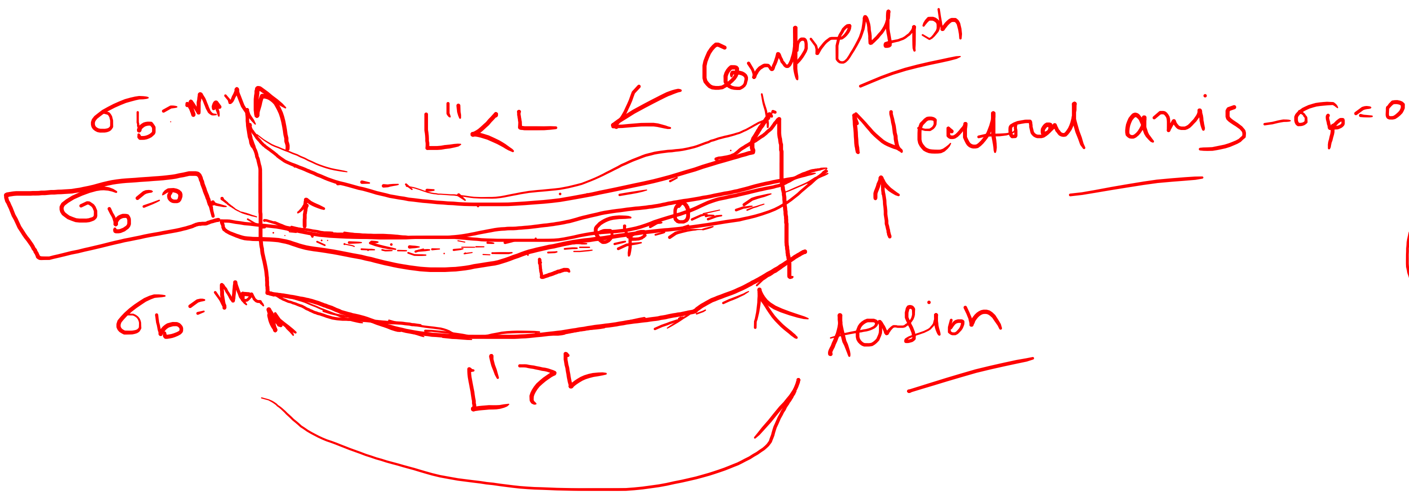
   Unequal forces



# Vertical bending of Chassis




Bending Equation

$\sigma_b$

Internal resistance force developed against force

# Vertical bending of Chassis

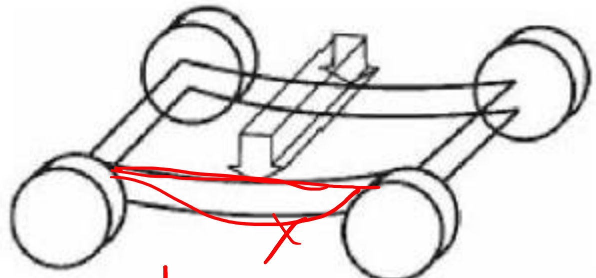
Material's utilization

I → Moment of Inertia

I' → Area of moment of Inertia

❖ **Vehicle components and payload apply downward forces**

➢ The central portion of the chassis tends to **sag downward**



Bending equation

$$\boxed{\frac{\sigma_b}{y} = \frac{M}{I} = \frac{E}{R}}$$

I

d, $y_{NA}$

$y = 0$, $y = d/2$

Where:

$\sigma_b$ = Bending stress: Internal resistance developed in the material to resist bending

M = Bending moment: How strongly the load tries to bend the chassis

I = Area moment of inertia: How effectively material is distributed away from the neutral axis

y = Distance from neutral axis: How far a material point is from the neutral axis

Z = I/y = Section modulus: Ability of cross-section to resist bending stress

E = Modulus of elasticity: Material resistance to elastic deformation (How stiff the material)

R = Radius of curvature after bending

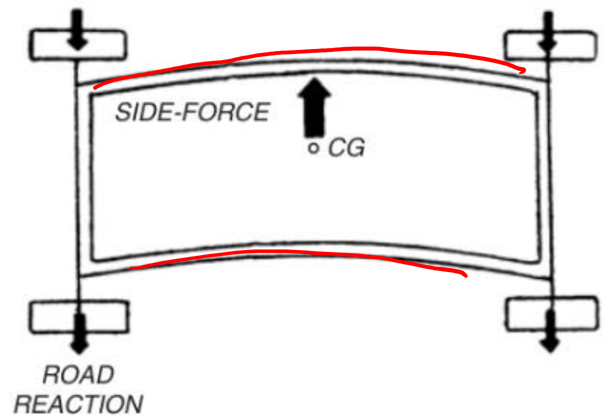
W

**Bending stiffness (EI): How difficult to bend the member**

**Design focus: Higher section modulus, Lower bending stress**

# Lateral Bending of Chassis

❖ **Two counter act forces: tyre-road contact points, while the vehicle mass has inertia acting through its centre of gravity (CG).**




➢ Mainly occurs when the vehicle is:

- Distortion of chassis members
- Change in wheel alignment
- Poor steering response
- Reduced handling precision
- Increased fatigue of structural joints

➢ For a vehicle negotiating a turn, Newton's second law applied to circular motion

C.F

$$F_y = \frac{mv^2}{R}$$

$\mathbf{F_y}$ = Net lateral force acting on the vehicle during cornering
**m** = Vehicle mass, **v** = Vehicle speed, **R** = Turning radius

**High lateral/bending stiffness**
**Less structural deformation - Better handling and durability**

# Longitudinal Torsion of Chassis

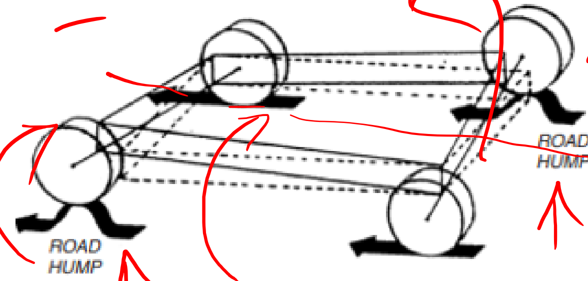
innovate achieve lead

❖ **Twisting action along the longitudinal axis of the chassis.**

θ - twisted

T Twist T

**Front view**




**Torsional Stiffness Equation**

$$K_t = \frac{T}{\theta}$$

Where:

$K_t$ = Torsional stiffness: Chassis' resistance to twisting

**T** = Applied torque: How strongly the external forces try to twist the chassis

**θ** = Angular twist: Amount by which the chassis actually twists

F = K x

Stiffness

M ← very bendy

T → long

**Design focus: Higher $K_t$ - Smaller chassis twist**

$T = K_t \theta$

$K_t = T/\theta$

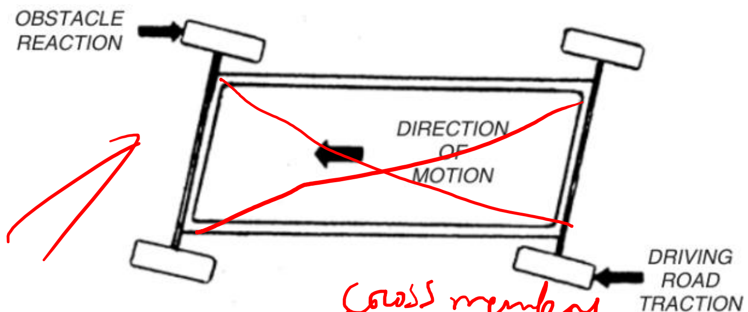
# Horizontal lozenging

- Distortion of the chassis from a rectangular shape toward a parallelogram/diamond shape when viewed from above.

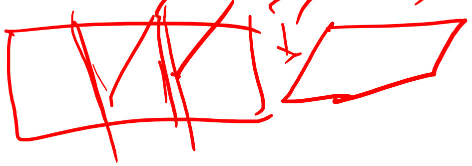
$$F_{brake,left} \neq F_{brake,right}$$

- **Hard braking on a split-friction road**
- ✓ Left wheels: high-friction dry road
- ✓ Right wheels: low-friction wet/icy road

**Top view**




Cross members



**Design Focus:**

**Strong cross-members, Box/closed sections, and Properly designed joints**

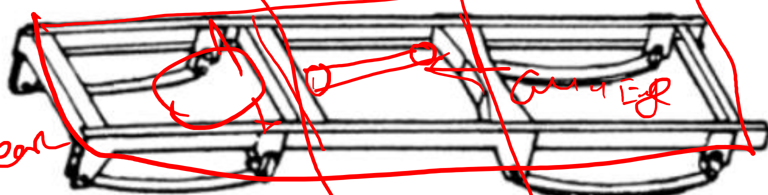
# Comparison

| Loading | Physical Meaning | Typical Cause |
|---|---|---|
| **Vertical Bending** | Chassis sags vertically | Vehicle weight + payload |
| **Longitudinal Torsion** | Chassis twists along its length | One wheel encounters a bump |
| **Lateral Bending** | Chassis bends sideways | Cornering |
| **Horizontal Lozenging** | Rectangular frame becomes skewed/diamond-shaped | Unequal braking/traction forces |

# Chassis Frame Design

Side rails, Cross rail

Cross members

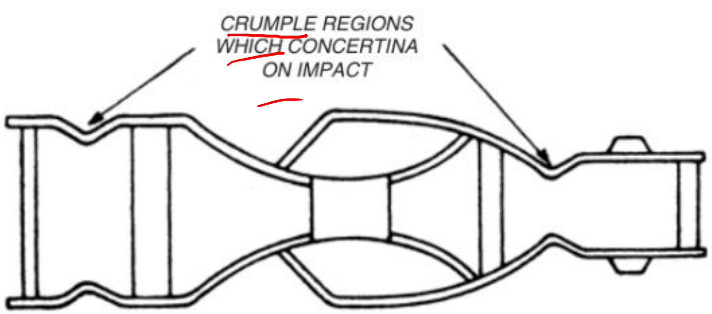


Ladder type frame

❖ **Ladder type frame:**

- Side members connected by cross-members.
- Side members carry most of the longitudinal bending loads.
- Cross-members connect the rails, maintain spacing and improve lateral/torsional rigidity.
- Cross-members are strategically placed near high-load locations, such as suspension, engine and transmission mounting regions.

$E = \frac{1}{2} m v^2$




Energy absorbing frame

❖ **Energy-Absorbing Frame**

- Excessive rigidity causes high crash deceleration.
- Crumple zones deform to absorb impact energy.
- Passenger compartment remains strong and protected.
- Reduces occupant injury.

$\frac{1}{2} m v^2 \to 0$, $= F_{avg} \times d$

$\boxed{F_{avg} \propto 1/d}$ $\Delta t \uparrow$, $F_{avg} \downarrow$

$\frac{1}{2} m v^2 = F_{avg} \times d$, $F_{avg} = \frac{\frac{1}{2} m v^2}{d}$

# Energy-Absorbing Frame

- Before impact, the vehicle possesses kinetic energy:

  $KE = ½mv^2$

- During collision, the crumple zone deforms and absorbs this energy:

  $½mv^2 \approx F_{avg} \times d$

- From impulse - momentum:

  $$\boxed{\begin{aligned} F_{avg} \times \Delta t &= m\Delta v \\ F_{avg} &= m\Delta v / \Delta t \end{aligned}}$$

$\downarrow F_{avg} \propto \Delta t \uparrow$

$mv \quad t_2 - t_1$

$mv_2 - mv_1$

$m[v_2 - v_1]$

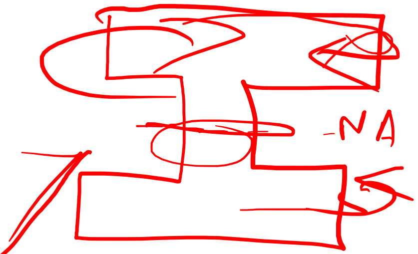
$m\Delta v$

where:
**m** = Vehicle mass, **v** = Impact velocity
**d** = Crumple/deformation distance
**Favg** = Average impact force
**Δv** = Change in vehicle velocity
**Δt** = Collision/stopping time

**Crumple Zone: Controlled Deformation**
**Increased Stopping Distance/Time: Reduced Impact Force**

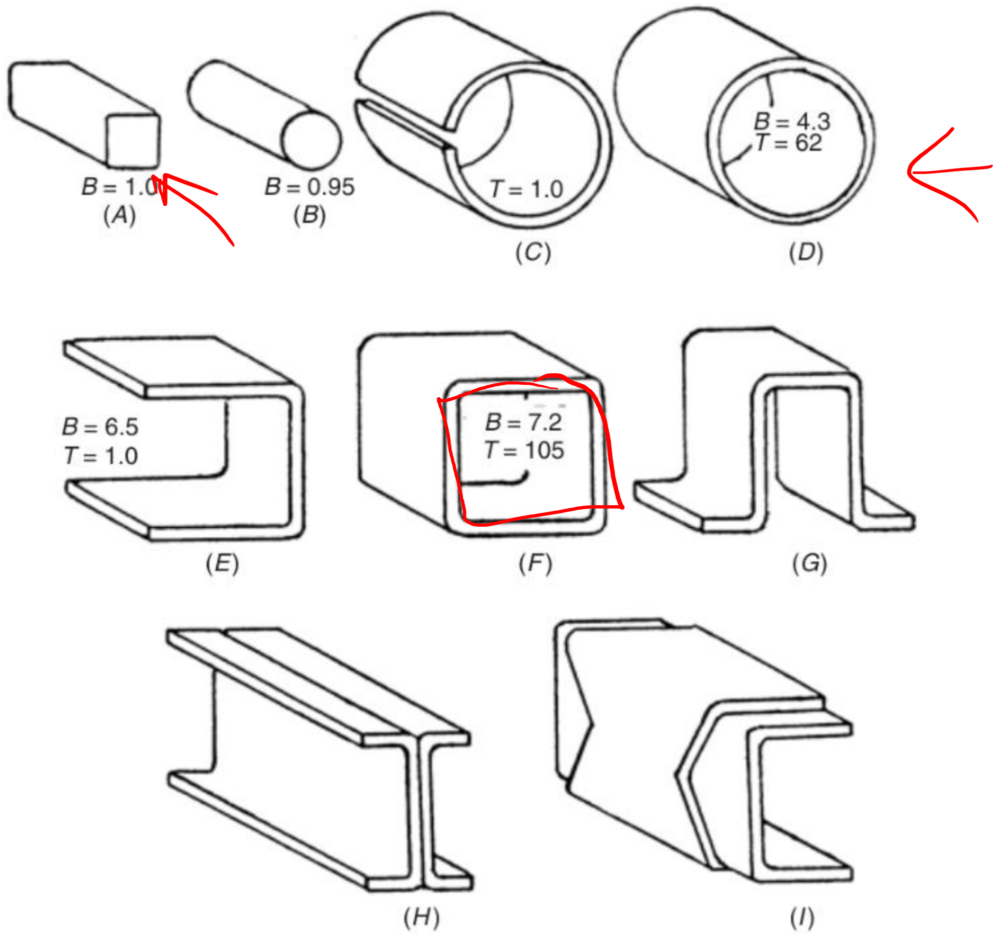
# Chassis Frame Sections

- During driving, chassis members experience **bending and torsional loads**.
  - Common cross-sections are:
    - **Solid sections** - round or rectangular.
    - **Closed hollow sections** - tubular or rectangular box sections.
    - **Open sections** - C, I or top-hat sections.




| Section | Bending | Torsion | Weight efficiency |
|---|---|---|---|
| Solid | Good | Good | Poor/heavy |
| C / I open | Very good | Poor | Good |
| Box / tube closed | Very good | Very good | Very good |

# Chassis Frame Sections




**A. Square solid bar | B. Round solid bar | C. Slotted circular tube | D. Closed circular tube | E. C-section | F. Rectangular box section | G. Top-hat section | H. I-section | I. C-channel with flitch plate**

# Numerical

$$\text{Load} = \frac{0.48 \times 1350 \text{ kgf}}{2}$$

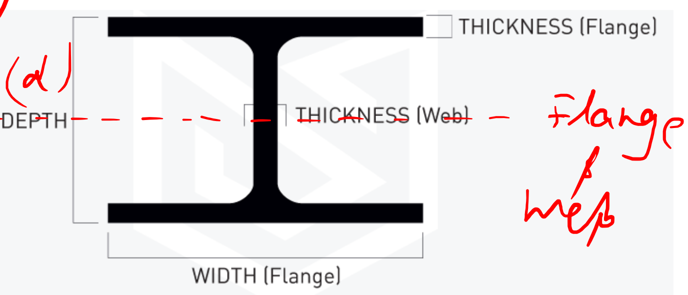
front

❖ **Q 1:** The load distribution between the front and the rear axle of a motor vehicle weighing 1350 kgf is that 48% of the total load is taken by the front axle. The width of the track is 140 cm and the distance between the centers of the spring pads is 66 cm.

$$\frac{\sigma_b}{y} = \frac{M}{I} \qquad 324 \times 37$$

➢ Design a suitable I-section for the front axle

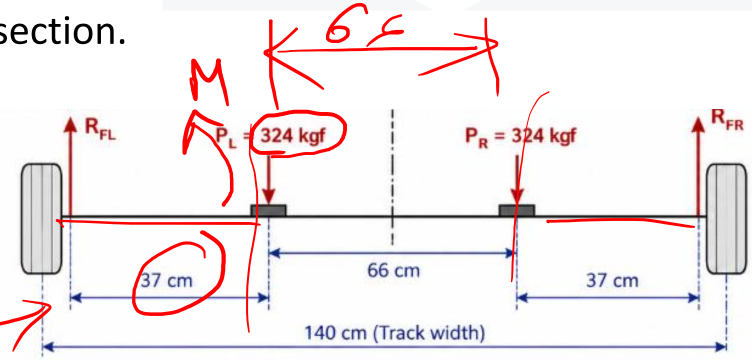
$d/2$ NA




✓ Assume

- Flange Width: 0.6 times overall depth of the section.
- Flange Thickness: 0.2 times overall depth of the section.
- Web Thickness: 0.15 times overall depth of the section.
- Assume a working stress of 915 kgf/cm². $\sigma_p$
- [Use consistent units: kgf and cm]

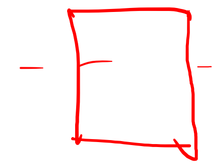
width, flange, web

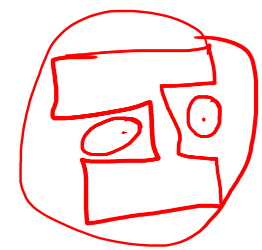



$$\frac{140 - 66}{2}$$

# Solution

- **Given:**
  - Vehicle weight, W = 1350 kgf
  - Front axle load = 48% of total load
  - Track width = 140 cm
  - Distance between spring pads = 66 cm
  - Working stress, $\sigma_b$ = 915 kgf/cm²





- **Front Axle Load:** Wf = 0.48 × 1350 = **648 kgf**
- **Load at each spring pad:** P = 648/2 = **324 kgf**
- **Distance between Wheel and Spring Pad:** a = (140 - 66)/2 = **37 cm**
- **Maximum Bending Moment:** Mmax = P × a = 324 × 37 **= 11,988 kgf·cm**

❖ **I-Section Dimensions:**

- Let overall depth = **d cm**
- Flange width = **0.6d**
- Flange thickness = **0.2d**
- Web thickness = 0.25 × 0.6d = **0.15d**

$\frac{bd^3}{12}$

# Solution

❖ **Using the bending equation:**

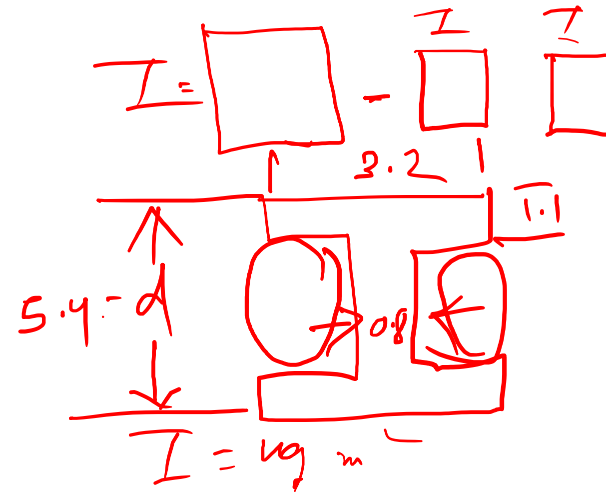
➢ $\boxed{\mathbf{M/I = \sigma_b / y}}$

where,
M = Maximum bending moment = **11,988 kgf·cm**
$\sigma_b$ = Allowable bending stress = **915 kgf/cm²**
y = Maximum distance from neutral axis = **d/2**

Area

❖ **Moment of Inertia of I-Section:**

**I = 0.0419d⁴ cm⁴**



**d = 5.4 cm**
**Flange width = 0.6d = 3.2 cm**
**Flange thickness = 0.2d = 1.1 cm**
**Web thickness = 0.15d = 0.8 cm**

# Appendix: Calculation of Moment of Inertia of I-Section

innovate achieve lead

❖ **Moment of Inertia (I) of I-Section**

➢ **Given Section Dimensions:**

Overall depth = d

Flange width = 0.6d

Flange thickness = 0.2d

Web thickness = 0.25 × 0.6d = **0.15d**

➢ **Consider I-section as:**

**I-Section = Outer Rectangle –**
void
**Two Empty Side Rectangles**

➢ **Outer Rectangle:**

Width, B = 0.6d

Depth, D = d

$I\_outer = BD^3/12$

$I\_outer = (0.6d)(d^3)/12$

$\mathbf{I\_outer = 0.05d^4}$

➢ **Empty Portion:**

Height of empty portion:

h = d − 2(0.2d)

h = d − 0.4d

**h = 0.6d**

➢ Combined width of two empty portions:

b = Flange width − Web thickness

b = 0.6d − 0.15d **= 0.45d**

➢ **Moment of Inertia of Empty Portion:**

$I\_empty = bh^3/12$

$I\_empty = (0.45d)(0.6d)^3/12$

$\mathbf{I\_empty = 0.0081d^4}$

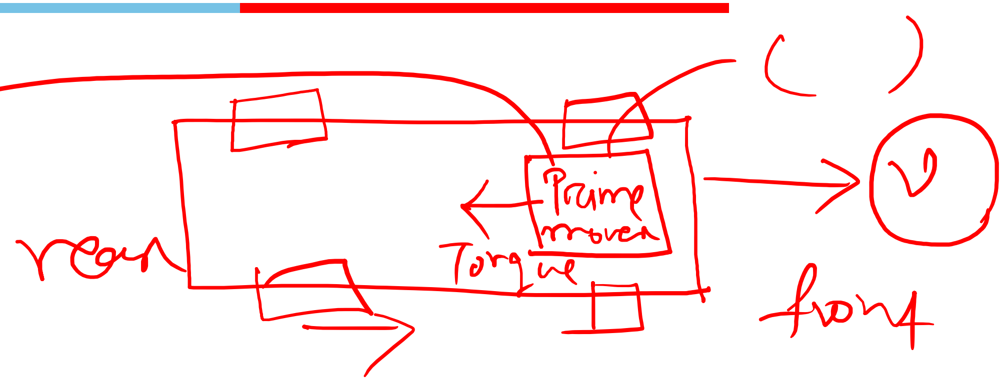
➢ **Moment of Inertia of Actual I-Section:**

I = I_outer − I_empty

$I = 0.05d^4 - 0.0081d^4$

$\boxed{\mathbf{I = 0.0419d^4\ cm^4}}$

$I = 0.0419d^4 cm^4$ ✓

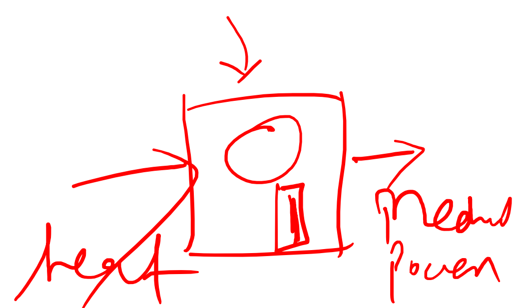



# IC Engine Fundamentals

# IC Engine Fundamentals

- Fundamentals of IC Engine
- Fuel conversion techniques
- Torque characteristics
- Emissions in engines

Heat Engine




H -> Enthalpy

U -> I.

Fuel + Air

$C_nH_m + O_2 \to CO_2 + H_2O$

External Combustion Engine -> Steam engine

Internal Combustion Engine

# Prime movers

❖ Stored energy to power: Mechanical rotation and vehicle propulsion.

| Prime Mover | Energy Source | Energy Conversion |
|---|---|---|
| **Internal Combustion Engine** | Petrol, CNG, $H_2$, Diesel | Chemical to Thermal to Mechanical |
| **Electric Motor** | Battery | Electrical to Mechanical |
| **Fuel-Cell + Motor** | Hydrogen | Chemical to Electrical to Mechanical |
| **Hybrid Powertrain** | Fuel + Battery | Chemical and Electrical to Mechanical |

hydrogen

Torque

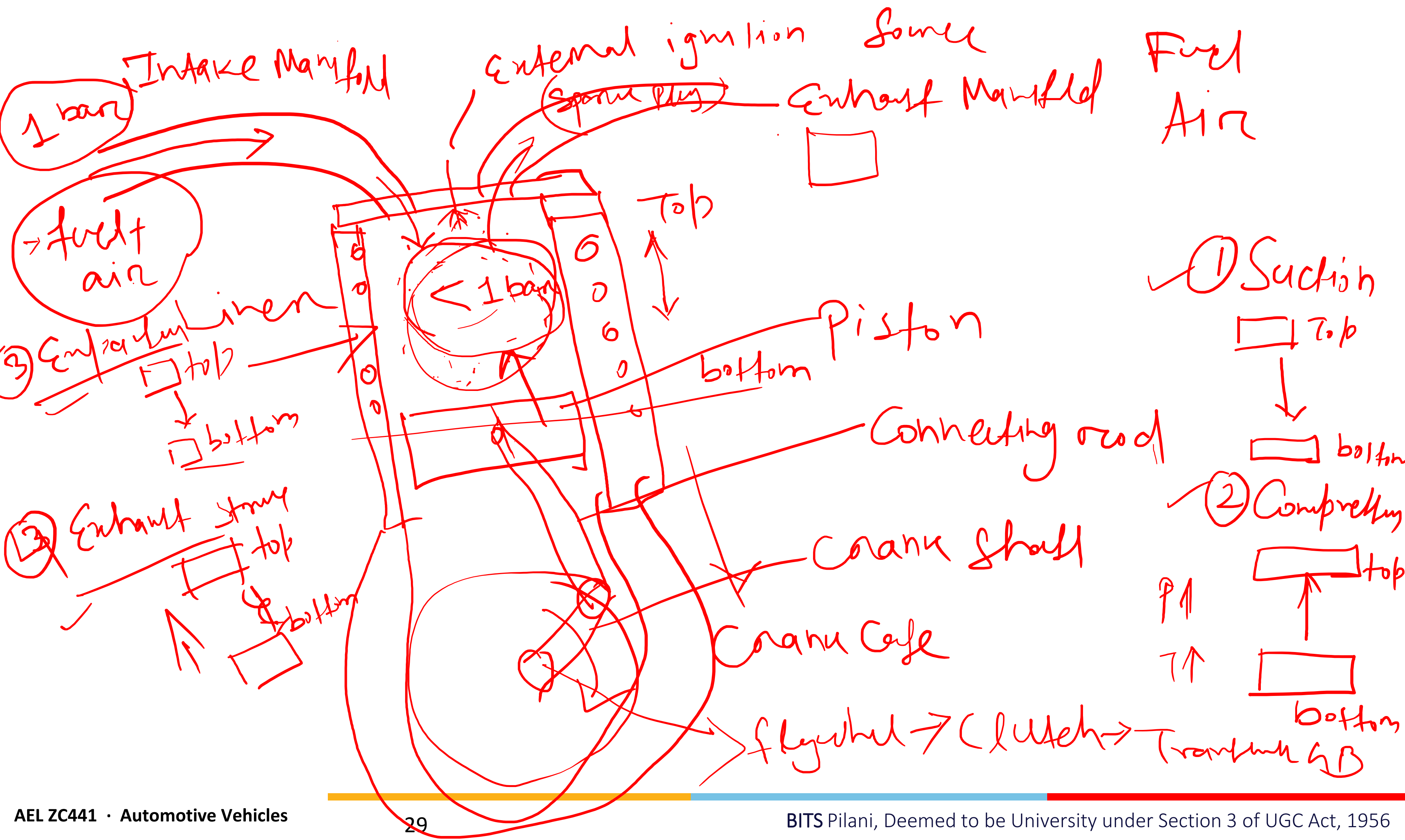
IC Engine

Motor +

Engine powering wheels

Eng

# Internal Combustion Engine



# IC Engine: Classification

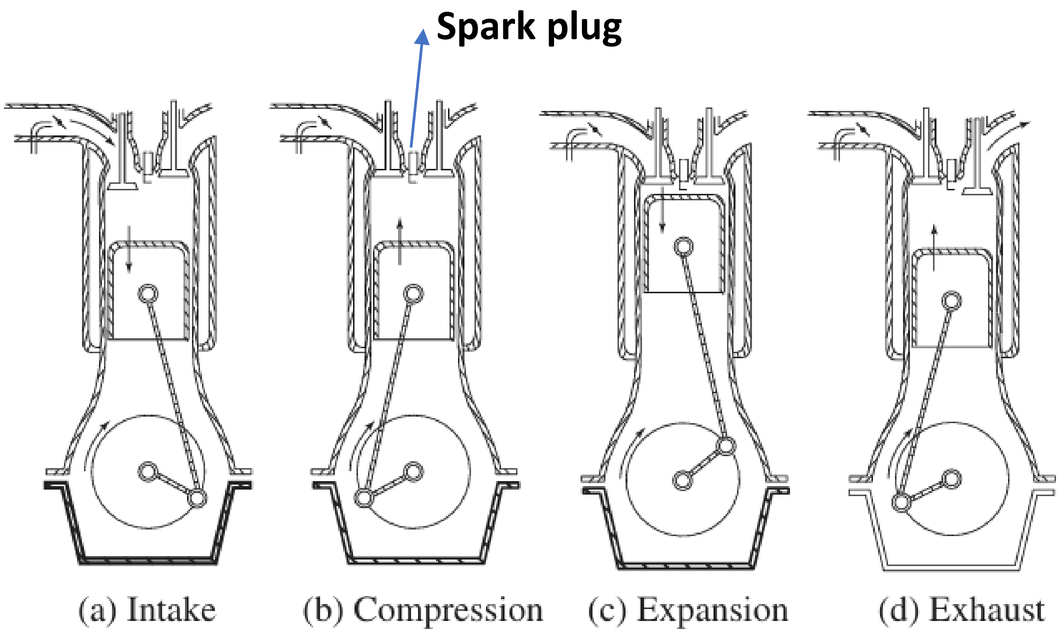
## 1. Working Cycle

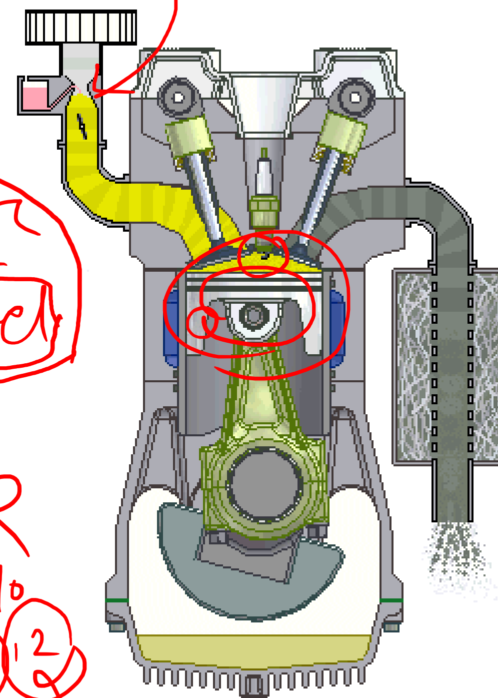
- **Four-stroke cycle engines:**
- **Two-stroke cycle engines:**

## 2. Method of Ignition

- **Spark ignition (SI)**
- **Compression ignition (CI)**

# Four-stroke Spark-Ignition Engine: Working



Air + fuel

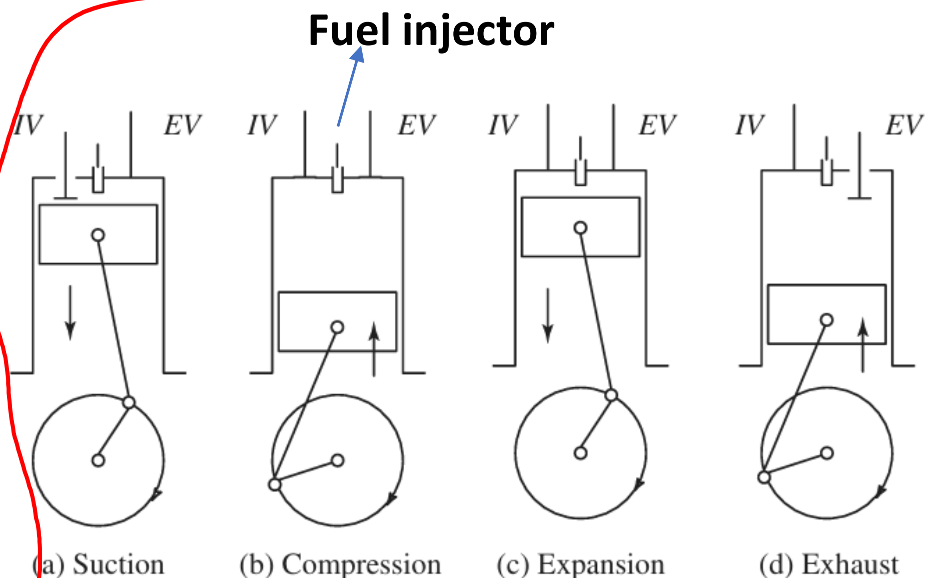
CR
9/10
11/12

**Figure: Working cycle of four-stroke SI engine**

Premixed (Air+fuel) Combustion Concept

Petrol Engine

# Four-Stroke Compression-Ignition Engine: Working




Working cycle of four-stroke Compression-Ignition engine

took fook

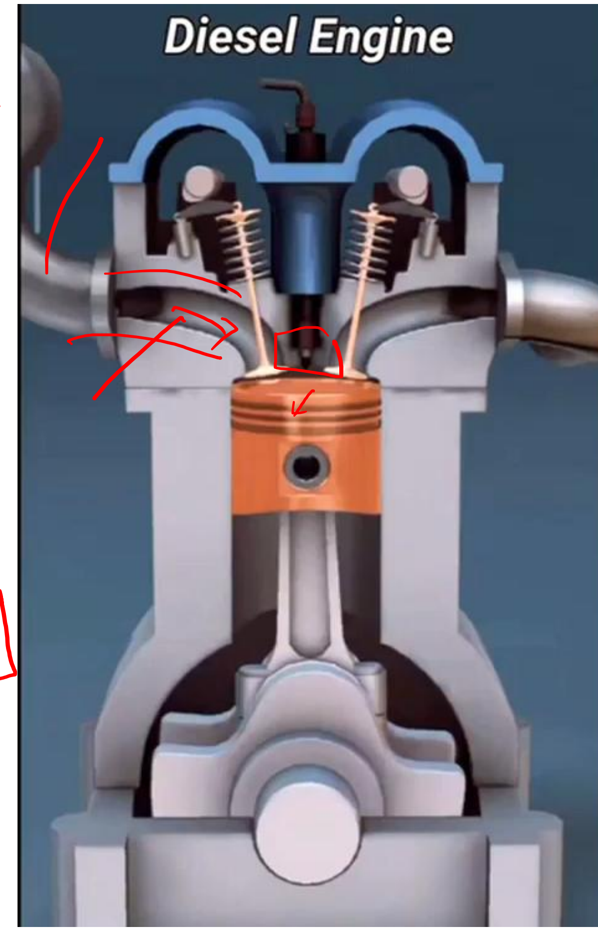
16, 18, 20, 22

CR?

Air

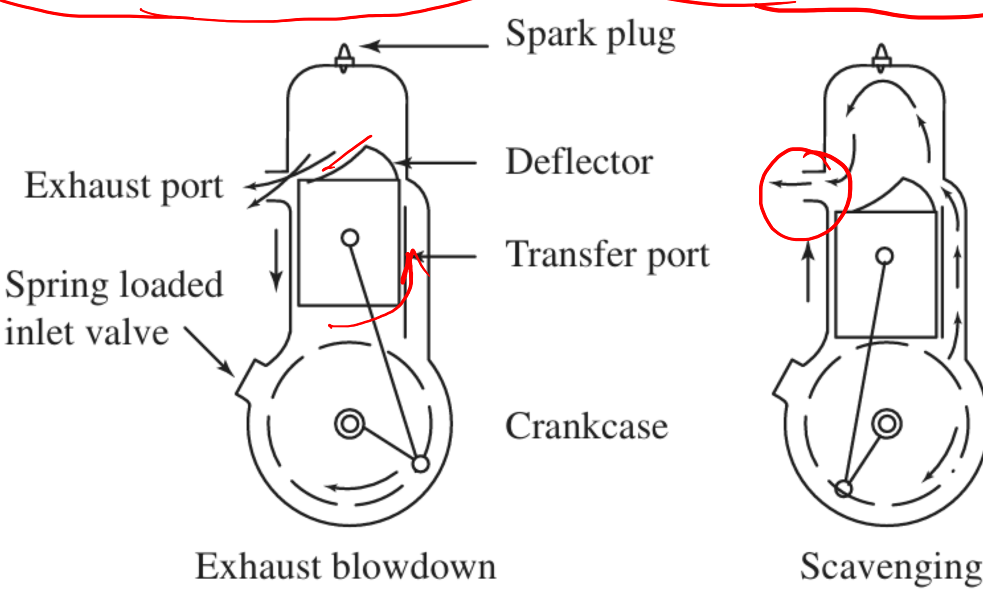
diffusion combustion concept

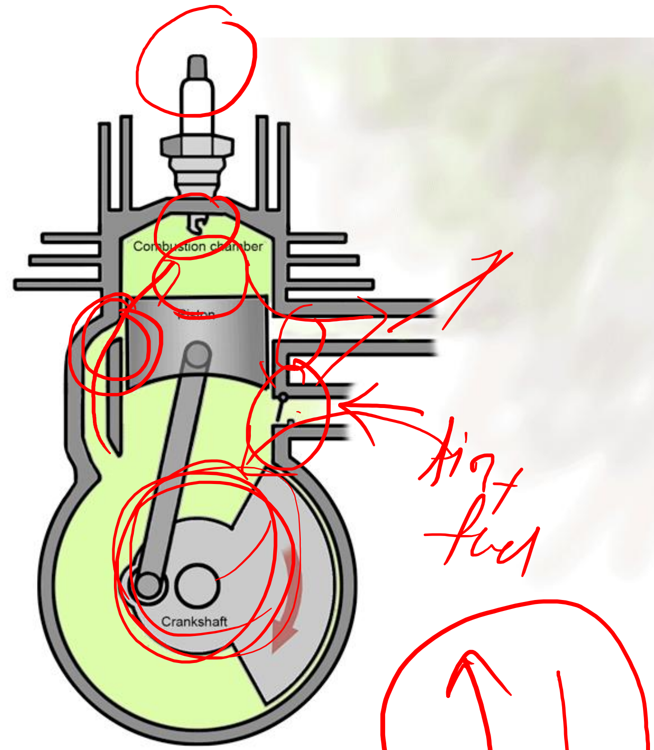
is there a throttle present in the diesel/CI cycle?



# Two-Stroke Engine: Working

Suction, Compression, Expansion, Exhaust




air + fuel

**Figure: Working cycle of 2-stroke engine**

2-cycle